

DECENTRALIZED NATIVE BOT SAVINGS VAULT

CHRONO

Time-locked. Temptation-proof.

A decentralized, mathematically enforced savings vault for native BOT on the BOT blockchain. Eliminates emotional trading and panic selling by locking capital into sovereign onchain positions with zero admin override capability.

0%	100%	128,000	968
ADMIN BACKDOORS OR OVERRIDES	ONCHAIN STATE STORAGE	INVARIANT TEST VERIFICATION CALLS	LIVE ON BOT TESTNET

The Problem: The "Hyper-Liquid" Curse

24/7 liquidity enables financial innovation, but wreaks havoc on human psychology and wealth preservation.

PSYCHOLOGICAL TRAP

1. Instant Temptation & Paper Hands

Every second a wallet is liquid, crypto investors face impulsive trades, FOMO dips, and panic-selling. Without rigorous friction, rational long-term savings strategies dissolve in volatile market swings.

SECURITY RISK

2. Risky Yield & Upgrade Exploits

Conventional "staking" platforms expose users to impermanent loss, bridge hacks, and proxy upgrade risks. Users looking for a simple, reliable lockbox are forced to gamble their principal in complex DeFi protocols.

INSTITUTIONAL GAP

3. Missing Commitment Devices

Traditional banks trap funds behind bureaucratic delays and predatory early-exit fees. Web3 wallets, conversely, offer zero native self-control tools to lock capital away from oneself.

MACRO ECONOMIC

4. High Token Velocity on Native BOT

Unanchored native token supply increases market churn. Without native time-locking sinks, circulating supply remains perpetually vulnerable to speculative runs and dumping.

The Solution: Programmatic Finality

Chrono transforms emotional trading into guaranteed onchain savings via immutable Solidity contracts.

DEFAULT SETTING

🔒 Strict Mode

Absolute mathematical finality. Once created, no early withdrawal exists under any condition. Not by the user, not by hackers, not by any admin. Funds unlock strictly when block timestamp reaches unlock time.

EMERGENCY VALVE

⚡ Flexible Mode

Allows early exits when genuine emergencies arise, with a hardcoded 10% penalty fee. Payout is returned immediately, and the penalty pool fuels protocol maintenance.

ZERO FEE

💎 0% Exit Fee at Unlock

Upon reaching the maturity timestamp, 100% of the locked BOT is released directly to the owner without platform cuts, management fees, or withdrawal penalties.

Pure Onchain Purity

All positions, timestamps, labels, and amounts are stored directly inside the Ethereum Virtual Machine state on Bohr BOT Testnet. Zero IPFS nodes, zero indexing delays, and zero centralized backend servers.

Target Personas & Market Opportunity

Capturing multi-million dollar capital flows across retail, institutional, and DAO ecosystems.

💎 HODLers & Savers

Investors seeking to lock away tokens for 1 to 5 years, shielding their portfolio from impulsive decision-making and market hysteria.

🏛️ DAO Treasuries

Protocol foundations and DAOs seeking trustless, transparent milestone vesting and runway lockups without complex escrow overhead.

🎯 Goal Accumulators

Individuals building earmarked savings for major life milestones: home deposits, tax liabilities, weddings, and tuition funds.

🌐 BOT Ecosystem

Acts as a macro supply sink, permanently taking circulating BOT out of active sell liquidity and enhancing ecosystem stability.

Behavioral Finance Thesis

Nobel Prize-winning behavioral economic research demonstrates that humans willingly pay a premium for commitment devices that constrain their future self from succumbing to short-term temptation. Chrono brings this economic truth directly to the blockchain.

Engineered for Elegance & Precision

A bespoke modern interface designed around the visual motif of vaults and time.

⏱️ Real-Time Progression

Circular SVG progress rings dynamically render elapsed percentages from lock creation to maturity date, accompanied by active second-by-second countdown tickers.

🛡️ Role-Aware Action Cards

UI surfaces strictly what the smart contract permits: Top-Up and Extend while active, a prominent Withdraw button upon unlock, and a clear disabled badge for Strict locks.

🔗 Hash Routing & Inspection

Position lookup by ID with shareable link routing (`#/lock/:id`) enables trustless verification of any onchain lock by anyone across the ecosystem.

Automatic Clock Drift Calibration

Syncs with latest block timestamps every 4 seconds to eliminate discrepancies between client local time and validator block time, preventing premature withdrawal reverts.

Zero Injection Security

All user-submitted labels are strictly rendered as sanitized plain text, guaranteeing 100% protection against XSS, DOM injection, or malicious payload rendering.

Tokenomics & Solvency Invariant

Simple, airtight economics based on native BOT and mathematically proven solvency.

SOLVENCY GUARANTEE

Mathematical Invariant

At all times and across all states, the vault satisfies the strict solvency equation:

```
address(chrono).balance == totalLocked + penaltyBalance
```

Every single wei of native BOT is accounted for. BOT cannot be minted, burned, or trapped in the contract.

PENALTY MECHANISM

10% Early-Exit Ceiling Rounding

On flexible early exits, penalties use ceiling integer division:

```
penalty = (amount * penaltyBps + 9999) / 10000;
```

Guarantees that tiny deposits (even 1 wei or 11 wei) never suffer under-collateralization or round-down leakage. Net payout + penalty always strictly equals principal.

1,000 BPS
10.0% EARLY EXIT PENALTY RATE

120 SEC
MINIMUM LOCK DURATION (DEMO MODE)

5 YEARS
MAXIMUM CONTRACT LOCK DURATION

Military-Grade Security & Verification

100% test pass rate across unit, boundary, reentrancy, fuzz, and stateful invariant suites.

AUDITED STANDARDS

OpenZeppelin 5.x

Inherits battle-tested `ReentrancyGuard`. All value-transfer routines strictly follow Checks-Effects-Interactions (CEI) to neutralize reentrant call hijacking.

BOUNDARY VERIFICATION

Exact Second Precision

Verified with `vm.warp` down to single-second granularity: `unlockTime - 1` strictly reverts, and `unlockTime` cleanly succeeds.

ATTACK DEFENSE

Isolated Failure Domains

Malicious contracts that intentionally reject BOT transfers revert on their own withdrawal without trapping or harming any other depositor's capital.

Foundry Test Suite Benchmark

✅ **37 Unit, Boundary, and Fuzz Tests: Verified parameter bounds, label constraints, multi-position ownership, and forward extensions.**

✅ **1 Stateful Invariant Test: 128,000 automated calls across 256 runs executing interleaved create, top-up, extend, withdraw, and early exit operations with 0 reverts and 0 discards.**

Live Deployment on BOT Testnet

Successfully deployed and verified on Bohr BOT Testnet (Chain ID 968).

Contract Specification

Contract Address:	0x0EA3c89F9f9A1dCdE0e5966D2e596E5985f8de3b
Chain ID:	968 (BOT Testnet)
Compiler:	Solidity ^0.8.24 (Solc 0.8.24)
RPC URL:	https://rpc.bohr.life
Block Explorer:	https://scan.bohr.life

Deployment Constants

Deployer Account:	0x44c0B36FA1a67eec472fF0AB032eC066847AAE1d
Penalty Recipient:	0x44c0B36FA1a67eec472fF0AB032eC066847AAE1d
Penalty Rate:	1000 bps (10.0%)
Min Lock Duration:	120 seconds
Max Lock Duration:	157,680,000 seconds (5 years)

Broadcast Execution Receipt

Tx Hash:	0xd36006e94a5b1b80e1a933e662ad87c17176c11e5951716a51ed9ddf930b728c
Execution Status:	Success • Block Confirmed • Gas Used: 3,343,447

Competitive Landscape

Why Chrono offers a superior self-commitment primitive over existing financial products.

Feature Dimension	Chrono Vault	Yield & Staking Protocols	Traditional Banking (CDs)
Early-Exit Protection	**Strict (0% Exit) & Flexible Modes**	Usually liquid; high trading temptation	Penalties & bureaucratic red tape
Smart Contract Risk	**Minimal (Immutable, Non-upgradeable)**	High (Proxy upgrades, flashloans)	N/A (Centralized insolvency risk)
Admin Backdoors	**Zero (0% admin keys or emergency pauses)**	Multi-sig admin pauses & upgrades	Full corporate & government control
Self-Sovereignty	**100% Non-custodial Web3 Wallet**	Governance attack exposure	KYC required, freezing risk
Maturity Fees	**0% Protocol Fee at Unlock**	Variable performance & withdrawal fees	Account maintenance & wire fees

Roadmap & Ecosystem Vision

Building the foundational savings standard for the decentralized web.

PHASE 1 • Q4 2026 (LIVE)

Testnet Launch

- Core vault smart contract deployed.
- 38/38 Foundry unit, fuzz & invariant tests.
- Full-featured React + TypeScript dapp.
- Live chain synchronization & custom errors.

PHASE 2 • Q1 2027

Mainnet & Audit

- Formal third-party smart contract audit.
- Deployment on BOT Mainnet (Chain ID 677).
- Mobile app PWA & deep wallet browser UX.
- Community incentivized savings sprints.

PHASE 3 • Q2 2027

Ecosystem Expansion

- Chrono SDK for third-party dapps & DAOs.
- Collaborative social milestone savings.
- Cross-dapp goal tracking integrations.
- Automated treasury payroll lockboxes.

Lock your BOT today. Remove temptation. Build the future.

Frontend: http://localhost:5174 • **Explorer:** https://scan.bohr.life/address/0x0EA3c89F9f9A1dCdE0e5966D2e596E5985f8de3b